

MEGGIE CHENG

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WORK EXPERIENCE

Tesla, Robotics Software Engineer, Python, C++

Apr 2026 – Present

Optimus humanoid robot (Tesla AI) – Scenario Evaluation & Simulation

- ❖ Engineered high-value simulation scenarios to validate new and existing Optimus features, advancing the validation pipeline toward operational autonomy.
- ❖ Built scenario creation, automation, and reporting tools that streamlined validation and improved simulator reliability and scalability.
- ❖ Performed functional, stress, integration, and performance testing across software-in-the-loop (SIL) and hardware-in-the-loop (HIL) systems.
- ❖ Designed test strategies for new features and triaged automated-test and bug-report failures, partnering with Simulation and Autopilot developers to fix and validate software changes.

Amazon Game Studios, Gameplay Engineer (Software Development Engineer I, II), C++

Aug 2022 – Jan 2026

New World Aeternum – Open world Massive Multiplayer Online game on PC, Xbox, PS5

- ❖ Shipped original PC and Aeternum console release, 3 expansions and 8 seasons.
- ❖ Designed and shipped 3 large-scale headline features end-to-end (architecture, backend integration, console support, optimization): Influence Races, a 600-player open-world PVP mode that held hub CPU under 85% at peak; Gear Set Storage loadouts, which drove over \$400k revenue in 2 months (purchased by 90%+ of MTX customers); and an MTX Store revamp that lifted store engagement 50%.
- ❖ Owned 17+ gameplay systems (items, loot, game events, store, etc.) as the go-to expert for knowledge and bug-fixing.
- ❖ Supported 7+ gameplay and backend teams: synchronized currency for the Catacombs mode, a furniture animation limiter for performance, and matchmaking race-condition fixes for async service calls.

PROJECTS

Cosmic Fruit Loops, Godot

GMTK Game Jam 2025 (4 days)

- ❖ A 2d puzzle game. Did art, UI design/programming, project management. Created with 3 teammates.
- ❖ Top 2.3% rated and top 11.7% most popular game in the entire jam out of 9,580 entries.

That Time I Got Reincarnated as a Gun and Had to Buy People to Protect My House, Godot

Pirate Software – Game Jam 16 2025 (2 weeks)

- ❖ A 3d tower defense game. Did art, game design, UI, project management. Created with 3 teammates.
- ❖ Top 16.5% most popular game in the entire jam out of 1,740 entries.

Pokédew Mystery Dungeon, Unity, C#

2022

- ❖ Designed and coded a 2D top-down rogue-like dungeon game with procedurally generated maps.

Monte Carlo Path Tracer, C++, OpenGL

2020

- ❖ Implemented a working ray tracer and path tracer following pbr-book.org that can render 3D scenes.

Mini Minecraft, C++, OpenGL

2019

- ❖ Created a Minecraft-style open sandbox game with 2 teammates.
- ❖ Coded player physics and procedural generation (L-Systems, rivers, trees, caves).

EDUCATION

University of Pennsylvania

Aug 2018 – May 2022

- ❖ **Bachelor of Science: Computer Science & Computer Graphics** (Digital Media Design), GPA: 3.94

SKILLS

- ❖ Languages: C++, Lua, Python, Java, C#, GDScript, GLSL, HTML, CSS, Javascript, Ruby on Rails, SQL
- ❖ Game Engines: Lumberyard/O3DE, Godot, Unity, Unreal
- ❖ Skills: Perforce, Git, Visual Studio, Amazon Q / Claude CLI (generative AI assistant), ImGui, CI/CD automation (Jenkins), distributed and event-driven systems, persistence, replication, server/client architecture, REST APIs, JSON/XML, scale testing, bot scripts and automated testing, OpenSearch Dashboards telemetry